

Customer Journey Map

5. PROJECT DEVELOPMENT PHASE

Date	4 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	3 Marks

Coding & Solution Overview

This document presents a code inventory and summarizes the core implementation behind the Human Development Index (HDI) Predictor system.

✂ Project Code Inventory

The solution is split into modular scripts:

1. [config.py](#): Encapsulates environment configurations, directories setup, and paths to raw and processed datasets.
2. [train_model.py](#):
3. Downloads raw UNDP datasets.
4. Runs preprocessing to melt indicator matrices into a long tidy table structure.
5. Generates Seaborn/Matplotlib visualization plots.
6. Standardizes features and trains a multi-variate Linear Regression model.
7. Serializes model and standard scaler.

8. [app.py](#):
 9. Configures Flask server routes, error overrides, and auth cookies.
 10. Intercepts unauthenticated dashboard requests, redirecting them to /login.
 11. Serves static metrics and handles predictions form POST.
 12. Runs StandardScaler standardization and predicts index values.
 13. Determines HDI categories and recommends policy actions.
 14. [script.js](#):
 15. Syncs sliders with input forms.
 16. Manages light/dark mode storage preferences.
 17. Drives country select prefills.
 18. Persists client histories and processes CSV exports.
 19. [style.css](#):
 20. Contains HSL-based colors mapping Light and Dark themes.
 21. Applies glassmorphism design tokens (backdrop-filter).
 22. Configures printer layouts to hide sidebars when printing report cards.
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Source Code Accessibility

All active code files are located within the root project workspace at C:\Users\User\.gemini\antigravity\scratch\hdi_predictor. You can browse and edit them directly. The automated unit tests in [tests/](#) verify model training outputs and Flask routes.